ECON 301 — Week 6: Heteroskedasticity

Introductory Econometrics · Lecture notes · Week 6 of 14

1. Where we are

In Weeks 1–5 we built the multiple linear regression model and the Gauss–Markov assumptions MLR.1–MLR.5. Under MLR.1–MLR.4 the OLS estimators are unbiased; adding MLR.5 (homoskedasticity) makes OLS the best linear unbiased estimator (BLUE) and justifies the usual standard errors.

2. The assumption we now relax

> MLR.5 (homoskedasticity): $\text{Var}(u \mid x_1, \ldots, x_k) = \sigma^2$. The error variance is the same for every value of the regressors.

Heteroskedasticity means the conditional variance of the error depends on the regressors: $\text{Var}(u_i \mid x_i) = \sigma_i^2$. A classic example is income and consumption: households with higher income have more room for discretionary spending, so the spread of consumption around the regression line grows with income.

3. Consequences for OLS

Heteroskedasticity does **not** cause bias or inconsistency in the OLS coefficient estimates — unbiasedness only needs MLR.1–MLR.4. R^2 and adjusted R^2 are also unaffected as measures of fit.

What breaks is inference. The usual formula $\text{Var}(b_1) = \sigma^2 / \text{SST}_x$ is derived under homoskedasticity, so the usual OLS standard errors are biased. As a result the t statistics, F statistics and confidence intervals based on them are no longer valid, even in large samples. OLS is also no longer BLUE: a more efficient linear unbiased estimator exists.

> Key exam point: heteroskedasticity leaves the coefficients unbiased but makes the usual standard errors — and therefore t and F tests — unreliable.

4. Heteroskedasticity-robust inference

White (1980) showed how to estimate $\text{Var}(b_1)$ without knowing the form of heteroskedasticity. The heteroskedasticity-robust variance estimator replaces σ^2 with the squared OLS residuals $\hat{u}_i^2$ observation by observation:

$$\widehat{\text{Var}}(b_1) = \sum (x_i - \bar{x})^2 \hat{u}_i^2 \,/\, \text{SST}_x^2$$

Its square root is the robust (White, or "HC") standard error. Robust t statistics are computed in the usual way but with robust standard errors. They are valid in large samples whether or not the errors are homoskedastic, which is why many applied economists report them by default (in Stata: `regress y x, robust`).

5. Testing for heteroskedasticity

Breusch–Pagan test. Estimate the model by OLS and save the squared residuals $\hat{u}_i^2$. Regress $\hat{u}_i^2$ on all regressors $x_1, \ldots, x_k$. Under the null of homoskedasticity the regressors should not explain $\hat{u}_i^2$; test their joint significance with an F test or the LM statistic $n \cdot R^2 \sim \chi^2_k$.

White test. Regress $\hat{u}^2$ on the regressors, their squares and their cross-products (or, in the special version, on $\hat{y}$ and $\hat{y}^2$). Compare $n \cdot R^2$ with a χ^2 distribution. The White test detects more general forms of heteroskedasticity than Breusch–Pagan, at the cost of using more degrees of freedom.

> A large $n \cdot R^2$ (small p-value) is evidence of heteroskedasticity. Failing to reject does not prove the errors are homoskedastic.

6. Weighted least squares (WLS)

If the form of heteroskedasticity is known up to a constant, $\mathrm{Var}(u \mid x) = \sigma^2 \cdot h(x)$, we can restore efficiency. Divide every variable in the equation by $\sqrt{h_i}$; the transformed error has constant variance σ^2, so OLS on the transformed equation is BLUE. This is weighted least squares with weights $1/h_i$: observations with a larger error variance get less weight.

In practice $h(x)$ is usually unknown and is estimated (feasible GLS), for example by modelling $\log(\hat{u}^2)$ as a linear function of the regressors. If the variance model is wrong, WLS is still consistent, but robust standard errors should be used for the WLS estimates too.

7. Summary and practice

• Heteroskedasticity: Var(u | x) varies with x. • OLS stays unbiased and consistent, but the usual standard errors, t tests and F tests are invalid. • Use robust standard errors for valid inference. • Detect with Breusch–Pagan or White tests. • If the variance function is known (or estimated), WLS is more efficient.

> **Practice.** In a regression of food spending on income for 500 households, the Breusch–Pagan statistic is $n \cdot R^2 = 31.4$ with 1 degree of freedom. (a) What do you conclude? (b) Which standard errors would you report for the income coefficient, and why? (c) Is the OLS estimate of the income effect biased?

Next week (Week 7): multicollinearity and model specification.